

MOSKVA VDNH, Russian Federation

WMO: 276120

Lat: 55.8314N

Lon: 37.6221E

Elev: 514

StdP: 14.42

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-1.6	-12.1	2.9	-6.8	-6.4	4.0	-0.3	10.0	20.0	9.2	21.1	2.4	350	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	85.9	69.9	82.4	68.4	79.2	66.9	71.8	83.1	69.9	79.8	68.2	77.1	2.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	104.8	77.4	66.2	98.6	75.3	64.7	93.2	73.2	35.8	83.4	34.2	79.8	32.7	77.5	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.7	7.0	-12.1	88.8	6.1	4.2	-16.5	91.9	-20.1	94.4	-23.5	96.7	-27.9	99.8		
(5)			-12.4	74.1	6.1	1.9	-16.8	75.5	-20.3	76.6	-23.8	77.7	-28.2	79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	19.9	20.7	30.2	44.4	55.8	62.9	68.0	63.5	53.2	42.5	31.4	23.9		
	DBStd	18.95	11.10	11.50	8.18	8.96	8.80	6.75	6.24	6.59	6.99	7.59	9.12	10.80		
	HDD50	4291	934	821	614	215	43	1	0	43	253	558	809			
	HDD65	8244	1399	1241	1078	618	306	119	39	106	358	699	1007	1274		
	CDD50	1795	0	0	0	47	223	388	558	419	140	20	0	0		
	CDD65	272	0	0	0	0	21	56	132	59	4	0	0	0		
	CDH74	2200	0	0	0	6	189	459	1030	493	23	0	0	0		
	CDH80	595	0	0	0	0	40	105	303	146	1	0	0	0		
	WSAvg	2.8	3.1	3.2	3.3	3.0	2.7	2.5	2.2	2.1	2.1	2.7	3.2	3.2		
	PrecAvg	27.0	1.8	1.6	1.4	1.6	2.1	2.9	3.4	3.1	2.6	2.5	2.1	2.0		
(15) Precipitation	PrecMax	35.1	3.9	3.8	3.5	4.1	5.9	6.3	7.6	6.6	7.2	6.3	5.7	3.9		
	PrecMin	17.9	0.1	0.1	0.2	0.4	0.0	0.4	0.2	0.6	0.5	0.0	0.2	0.6		
	PrecStd	4.4	0.8	0.8	0.7	0.8	1.3	1.3	1.7	1.5	1.3	1.5	0.9	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	39.6	56.2	74.3	84.5	86.8	92.2	91.2	77.2	66.3	52.8	46.1		
		MCWB	38.0	36.8	46.3	58.5	67.4	68.5	72.2	72.5	64.6	57.6	50.1	44.3		
	2%	DB	36.6	37.5	48.7	68.7	79.8	83.0	86.6	83.7	72.0	59.9	48.0	41.8		
		MCWB	35.3	35.0	40.8	55.3	64.9	67.7	70.9	69.8	62.2	54.7	46.0	40.0		
	5%	DB	35.1	36.0	44.3	63.9	74.8	79.4	83.1	79.2	67.9	55.9	45.2	38.3		
		MCWB	33.8	34.1	38.3	52.7	62.2	66.4	69.3	66.9	60.2	52.2	43.3	36.8		
	10%	DB	33.5	34.8	40.7	59.8	70.8	75.6	79.7	75.4	64.2	52.8	42.4	35.8		
		MCWB	32.3	33.1	36.2	50.3	59.5	64.4	68.1	65.0	58.3	49.9	40.6	34.4		
	0.4%	WB	38.4	37.9	46.9	59.6	69.9	72.0	74.0	73.3	66.5	58.7	50.1	44.5		
		MCDB	39.0	39.0	55.3	72.4	82.8	83.2	85.9	86.1	74.6	63.5	52.6	45.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.5	35.7	42.0	56.2	66.1	69.4	72.5	70.8	63.2	55.5	46.3	40.1		
		MCDB	36.3	36.6	47.3	66.8	76.5	79.8	84.1	82.0	69.6	59.2	47.9	41.6		
	5%	WB	34.0	34.5	39.1	53.7	63.3	67.5	70.7	68.4	61.0	52.6	43.5	37.1		
		MCDB	34.9	35.6	43.1	62.0	73.3	77.2	80.6	77.3	66.3	55.4	45.0	38.2		
	10%	WB	32.4	33.3	36.8	51.1	60.8	65.5	69.0	66.2	58.8	50.2	40.7	34.6		
		MCDB	33.5	34.7	40.1	58.8	68.3	73.6	77.7	73.1	63.9	52.4	42.2	35.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	9.5	11.6	15.6	17.3	15.7	15.7	15.6	13.8	9.6	6.6	6.8		
		MCDBR	6.3	7.6	16.8	22.6	22.8	21.5	21.0	21.6	20.1	14.5	8.6	6.3		
		MCWBR	6.4	6.7	10.6	11.8	10.6	9.2	8.3	9.2	10.9	9.6	7.6	5.9		
		MCDBR	6.3	6.8	13.9	21.0	20.9	19.4	18.9	20.0	17.6	13.2	7.9	6.3		
	5% WB	MCWBR	6.5	6.5	9.4	11.7	10.3	9.4	8.6	9.4	10.3	9.4	7.4	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.338	0.353	0.427	0.406	0.392	0.440	0.445	0.394	0.362	0.327	0.291		
	taud		2.121	2.064	2.109	2.100	2.286	2.348	2.239	2.207	2.330	2.348	2.230	2.114		
	Ebn at Noon		184	220	252	249	265	270	253	242	239	216	174	155		
	Edh at Noon		19	30	37	44	38	36	40	39	30	23	18	16		
(44) All-Sky Solar Radiation	RadAvg		185	456	943	1270	1668	1736	1707	1375	875	399	154	99		
	RadStd		37	62	94	112	153	167	183	147	116	70	27	29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+1.28	N/A	N/A	N/A	N/A	+1.26	N/A	N/A	N/A	N/A	
(47) Regional (10 neighbors)	+1.31	N/A	N/A	N/A	N/A	N/A	-265	-394	+198	N/A	

Nomenclature: See separate page